Silverio Martínez Andrés Grupo 35 Tarea 7

9/10

Nombre:

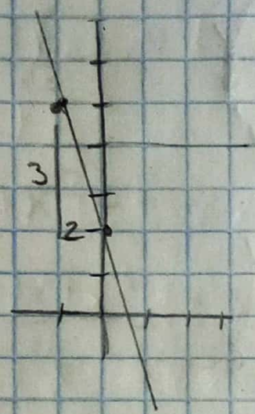
Tema: IS PO M

Día	Mes	Año	Folio
09	09	21	

Tarea 7

1.- Determinar dominio y rango de las siguientes funciones

a) $y = -3x + 2$



$D_f = \mathbb{R} = (-\infty, \infty)$

$R_f = \mathbb{R} = (-\infty, \infty)$

b) $y = -2 + \sqrt{5 - 4x - x^2}$

$y + 2 = +\sqrt{5 - 4x - x^2}$

$(y+2)^2 = 5 - 4x - x^2$

$x^2 + 4x + (y+2)^2 = 5$

$\frac{4}{2} = 2f$

$2^2 = 4c$

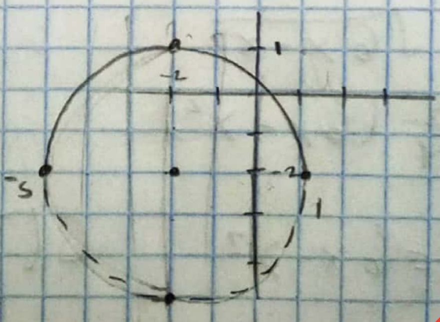
$x^2 + 4x + 4 + (y+2)^2 = 5 + 4$

$(x+2)^2 + (y+2)^2 = 9 \rightarrow$

Circunf $C(-2, -2)$

$r = 3$

Representa arriba de la circunferencia



$D_f = [-5, 1]$

$R_f = [-2, 1]$

Silverio Martínez Andrés Grupo 3S Tarea 7

Nombre:

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14	09	21	

Dominio son los valores que toma "x" y el rango los valores que toma y.

c) $y = 2\sqrt{4-x}$

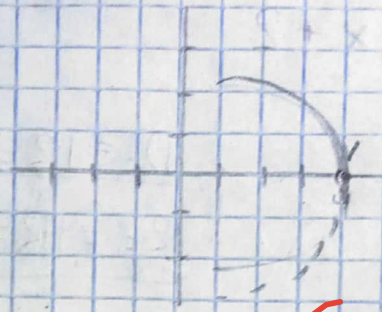
$y^2 = 4(4-x)$

$y^2 = 4(-1)(x-4)$

$y^2 = -4(x-4)$ f. ord

parábola "hor" abre izq

$V(4,0)$



$D_f = (-\infty, 4]$

$R_f = [0, \infty)$

d) $y = 1 - \frac{2}{3}\sqrt{6x - x^2}$

$y - 1 = -\frac{2}{3}\sqrt{6x - x^2}$

$(y-1)^2 = \frac{4}{9}(6x - x^2)$

$\frac{9}{4}(y-1)^2 = 6x - x^2$

$x^2 - 6x + \frac{9}{4}(y-1)^2 = 0$

$-\frac{6}{2} = -3$

$-3^2 = 9$

$x^2 - 6x + 9 + \frac{9}{4}(y-1)^2 = 9$

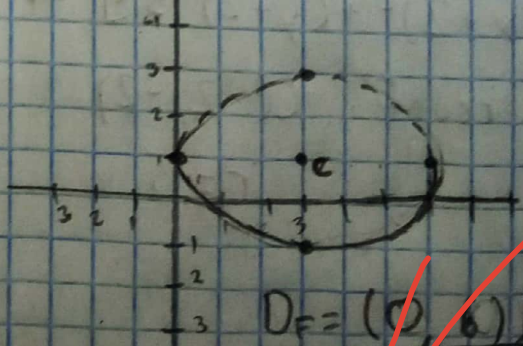
$(x-3)^2 + \frac{9}{4}(y-1)^2 = 9$

$\frac{(x-3)^2}{9} + \frac{\cancel{9}(y-1)^2}{\cancel{9}(4)} = \frac{9}{9}$

$\frac{(x-3)^2}{9} + \frac{(y-1)^2}{4} = 1$ f. ord

$C(3,1)$

$a = 3$
$b = 2$



$D_f = [0, 6]$

$R_f = [-1, 1]$

Silverio Martínez Andrés Grupo 35 Tarea 7

Nombre: | Día | Mes | Año | Folio

Tema:

e) $y = -\frac{3}{4}\sqrt{x^2-16}$ hip "hor" $C(0,0)$

$a=4, \quad b=3$

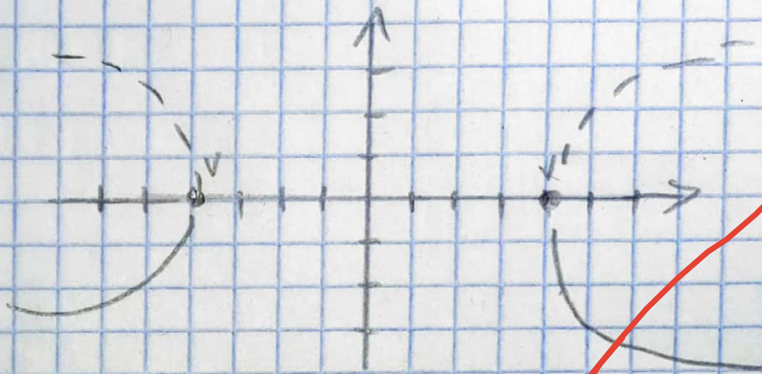
$y^2 = -\frac{9}{16}(x^2-16)$

$\frac{16}{9}y^2 = x^2-16$

$x^2 - \frac{16}{9}y^2 = 16$

$\frac{x^2}{16} - \frac{16}{16(9)}y^2 = \frac{16}{16}$

$\frac{x^2}{16} - \frac{16}{9}y^2 = 1$ F. ord



$D_F = (-\infty, -4] \cup [4, \infty)$

$R_F = (-\infty, 0] \cup [0, -\infty)$